

EE/CprE/SE 4920 STATUS REPORT 6

3/26/26 - 4/9/26

Group Number: sdmay26-40

Project Title: True Force Technologies Arcade Rack

Client: True Force Technologies

Advisor: Matt Post

Team Members/Role:

Jacob Garcia - Frontend Development

Dylan Longlett - Frontend Development, Electronics

Andy Drafahl - Team Manager, Frontend Development

Fadi Masannat - Frontend Development

Sofi Gutierrez - Frontend Development

Parnika Dasgupta - Client Liaison, Frontend Development

Colin Yuska - Electronics

Weekly Summary

This past couple of weeks, the team has continued on minigame development and cabinet repair. The Rocket game and UFO game are just about finished, and the Balloon Burst game is on its way as well. Repairs to the cabinet are still underway, and the team is working out some issues with the hat.

Past Week Accomplishments

Jacob Garcia

- Fix clouds
- Display text to keep lifting when user stops lifting in rocket launch stage
- Fix game selection screen
- General rocket launch bug fixes
- Game selection screen QoL changes
- Back button implemented across all games

Parnika Dasgupta

- Worked on setting up the background of my game
- Worked on adding animations to my game

Dylan Longlett

Fadi Masannat

I refined the default screen since it needed some UI adaptations, spacing and client requests to accommodate. I started making tests and finished the testing infrastructure architecture document that plans the testing pyramid and how to ratio our tests from unit to integration etc.

Sofi Gutierrez

I added logic so the screen is the same size of the physical screen, added "collecting cows" logic and score multiplier, along with changing the logic of the force-pulled movement and its range with the collecting cows.

Colin Yuska

Worked on backtracking to "pre-hat" electronics, started testing nominal voltages to then further test hats applicability. Researched new light specifications as per advisors suggestion.

Andy Drafahl

I created animations for the water balloons in Parnika's minigame, created alternate versions of avatar costumes, and worked on the code for Parnika's minigame.

Pending Issues

The cabinet still needs a lot of cleaning up, but we're making steady progress on it.

Individual Contributions

Name	Individual Contributions	Hours This Week	Hours Cumulative
Jacob Garcia	- Fix clouds - Display text to keep lifting when user stops lifting in rocket launch stage - Fix game selection screen - General rocket launch bug fixes - Game selection screen QoL changes - Back button implemented across all games	5	36*
Parnika Dasgupta	- Fixed background - Added animations - Struggled with problems on Github	4	18
Dylan Longlett			
Sofi Gutierrez	- Added features such as collecting cows - Changed the range and logic of player movement when collecting cows - Edited the score system - Added more visuals in the menu state	5	31
Colin Yuska	- Worked on getting electronics back into their original state for software testing. - Began testing nominal voltages of the pull sensor to report for Arduino hat testing. - Started researching new LED light solutions with advisor's recommendations.	6	21.38
Fadi Masannat	- Refined the UI of the main game as per client's requests - Setup the architectural plan for testing design and started making end to end tests	4	26
Andy Drafahl	- Created animations for red and blue water balloons expanding. - Created animations for red and blue water balloons bursting.	11.6	34.08

	- Created a logo for Balloon Burst. - Revamped title screen, avatar selection, and music playing for Balloon burst.		
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Plans for the Upcoming Week

Jacob Garcia

Finish up development for Rocket Launch.

Parnika Dasgupta

Work with Andy to figure out what he changed in my game, and also work with him to have the game running on the Arcade.

Dylan Longlett

Fadi Masannat

I still have to setup a GitLab CI pipeline that runs the tests using runners with every commit or PR. I want the testing to be complete by the end of next week. And harden the flow and states inside the app and start working on the end to end hardening of the codebase.

Sofi Gutierrez

I came in a couple of times to test my code, but the cabinet was being worked on, so I still have testing to do, and I was going to add the 3 letter player id to the beginning, like the other game, and include it in the leaderboard with a date stamp. I will also start writing test cases to test our code

Colin Yuska

Completely finish setting the cabinet back up, and assist with the hat testing so a new improved one can be ordered before the presentation date.

Andy Drafahl

I plan to finish up my parts of Parnika's minigame, then pass it off to her for gameplay logic, additional edits, and testing with the force rack. I also plan to add the avatars to the Powerpuller 3000 minigame and to keep working on music for all minigames.

Summary of Weekly Advisor Meeting

Matt pointed out that if we get the cabinet back together and running, he'll be able to come in and measure voltages. He raised a concern about the thermal protection that we removed from the Raspberry Pi, but we're working on a solution that allows us to use the hat and the Pi at the same time, as the case with the thermal protection gets in the way. We'll monitor the temperature in the meantime. We also decided to prioritize getting the hat running, implementing sound, mounting the LEDs, and cleaning up the cabinet in general, as the project deadline is quickly approaching.

